

Longitudinal Drive Motor, Coupling and Gear Box Replacement

[\(Click Here for Summary Version\)](#)

1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
2	Not Applicable	45 mins	10 mins

2 Overview

The procedure that follows is for replacing the Longitudinal Drive Motor, Coupling and Gear Box Assembly. The motor or assembly must be removed from the Magnet Room prior to replacement. The Flexible Coupling can be replaced by following either the motor or the gear box replacement procedure.

3 Preliminary Requirements

3.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
Nonmagnetic Service Tool Kit	1	-	5113258 or 5112581	-

3.2 Replacement Parts

Item	Qty	Effectivity	Part#	Manufacturer
Longitudinal Drive Motor	1	-	See FRU manual.	-
Flexible Coupling	1	-	See FRU manual.	-
16 mm Gear Box (Speed Reducer)	1	-	See FRU manual.	-

3.3 Safety



WARNING

**PERSONAL INJURY AND EQUIPMENT DAMAGE!
 STRONG MAGNETIC FIELD PRESENT.**

WHEN SERVICING ANY MAGNETIC EQUIPMENT, IT IS CRITICALLY IMPORTANT THAT THE SERVICE ENGINEER CONSCIOUSLY PLAN THE PATH TO BE TAKEN WHEN MOVING HIGHLY FERROUS DEVICES IN THE MAGNET ENVIRONMENT. THE PATH SHOULD BE AS FAR FROM THE MAGNET AS PRACTICAL AND AVOID HIGH FLUX-DENSITY FIELDS.

SAFETY REQUIREMENTS:

- **MOVEMENT OF FERROUS MATERIAL IN THE MAGNET ROOM *MUST* FOLLOW THE GE SERVICE PROCEDURE FOR THAT DEVICE. WHEN EXITING, MOVE AWAY FROM THE MAGNET IN THE MOST DIRECT MANNER POSSIBLE. EXCEPT WHEN MOVING FERROUS MATERIAL TO AND FROM ITS NATIVE LOCATION ON OR NEAR THE MAGNET, THE STATIC MAGNETIC FIELD IN ANY PORTION OF THE SERVICE PATH SHALL NOT EXCEED 200 GAUSS.**
- **TWO (2) MR SAFETY-TRAINED PERSONNEL *MUST* BE PRESENT AT ALL TIMES WHEN SERVICING HIGHLY FERROUS DEVICES IN THE AREAS OF MAGNETIC FIELDS.**

WHEN PLANNING A SERVICE PATH, IT IS **CRITICAL** THAT THE PATH BE CLEAR AND SUFFICIENTLY WIDE. ENSURE THAT THERE ARE NO TRIP HAZARDS, OBSTACLES, CLUTTER, SLIPPERY SURFACES OR OTHER ITEMS EVEN PARTIALLY RESTRICTING THE PATH. IF THERE ARE PORTABLE OBSTACLES IN A PATH, REMOVE THEM FROM THE AREA AND REPLACE THEM AFTER THE SERVICE ACTION IS COMPLETED. IT IS REQUIRED TO WALK THE PATH PRIOR TO BEGINNING SERVICE TO ENSURE THAT THERE IS SUFFICIENT SPACE THROUGH WHICH TO PASS FOR YOURSELF AND THE OBJECT BEING SERVICED.



CAUTION

Ferrous Material Hazard!

Longitudinal Drive Motor contains ferrous material.

Holding this assembly too close to the magnet bore will forcibly attract it to the magnet. To prevent possible bodily injury or material damage, remove the longitudinal drive motor and mounting bracket from the Magnet Room to install the new longitudinal drive motor. (See [Illustration 2](#) for proper process to remove motor from Magnet Room.) Two people are required at all times for this procedure.

3.4 Required Conditions

Condition	Reference	Effectivity
Confirm there is a clear path before performing any work.	-	-

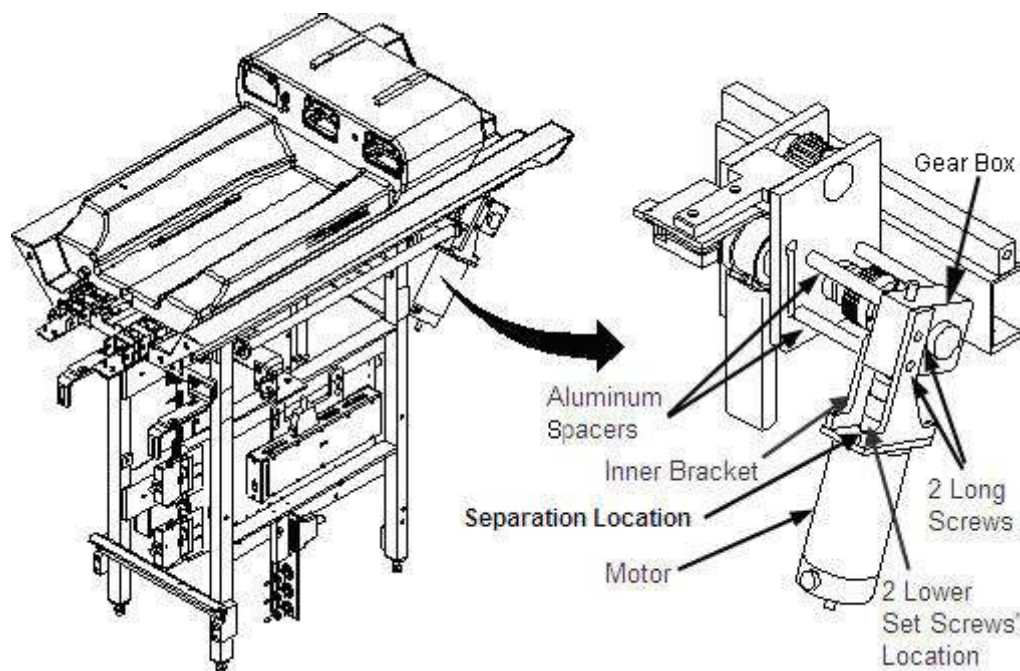
4 Procedure

4.1 Longitudinal Drive Motor Replacement

The Longitudinal Drive is located in the rear pedestal. Two people are required to replace the Longitudinal Drive Motor due to its strong attraction to the magnet. The motor must be removed from the Magnet Room prior to replacement.

1. Perform LOTO on the PDU. See the *MR Service Safety Manual*, PN 5452735.
2. Remove the rear pedestal left side cover (as viewed from the rear of the magnet).
3. Disconnect the cables from the motor.
4. Loosen the two lower set screws (about 90 degrees apart) on the motor shaft coupling shown below.

Illustration 1: Long Drive Motor Removal





CAUTION

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Holding this assembly too close to the magnet bore will forcibly attract it to the magnet. To prevent possible bodily injury or material damage, remove the longitudinal drive motor and mounting bracket from the Magnet Room to install the new longitudinal drive motor.

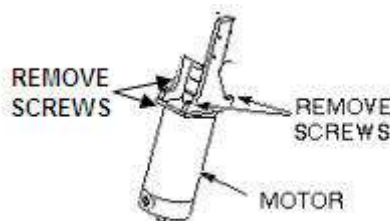
5. While one person holds onto the motor firmly with both hands, the other person should loosen and remove the two long motor mounting screws shown in [Illustration 1](#). (When removing the two long motor mounting screws from the bracket, the aluminum spacers may become free as well.)
6. Twist the motor/bracket assembly to free the inner bracket. Insert a screwdriver into the separation location shown in [Illustration 1](#) and pry apart.
7. With the last screw removed, carefully remove the motor from Magnet Room by placing the motor on the floor and against the wall farthest from the magnet.
 - a. Slide the motor to the front of the Magnet Room as shown in the illustration below until outside of the 200 Gauss area.
 - b. Retain the two long screws and the aluminum spacers for the replacement motor.

Illustration 2: Removal of Drive Motor from Magnet Room



8. With the motor **outside** of the Magnet Room, remove the four mounting screws (shown below) from the motor mounting bracket. (These screws go through the mount and secure the motor by pulling it tight to the mount.)

Illustration 3: Motor and Bracket Separation



9. Install the replacement motor onto the bracket, and reinstall the four mounting screws on the bracket as shown above. Do **not** completely tighten the mounting screws as some bracket movement is needed for reconnecting to the clutch assembly.

**CAUTION****Ferrous Material Hazard!**

Longitudinal Drive Motor contains ferrous material.

Holding this assembly too close to the magnet bore will forcibly attract it to the magnet. To prevent bodily injury or material damage, remove the longitudinal drive motor and mounting bracket from the Magnet Room to install the new longitudinal drive motor.

10. Carefully bring the motor and bracket assembly into the Magnet Room along the same path used to remove the assembly. (See [Illustration 2](#).)
11. To reinstall the motor and bracket assembly in the Rear Pedestal, ensure that a second person holds onto the motor firmly with both hands, while the first person reconnects the motor and bracket assembly to the clutch assembly with the two long screws and the aluminum spacers.

**NOTICE**

Be sure to apply Loctite 242 to all screws used to reassemble the replacement assembly, including the set screws in the clutch.

12. Draw the four screws of the bracket up evenly, tightening the motor in its mount.
13. Tighten the two set screws on the coupling to finish the motor installation.
14. Reattach the cables to the motor and the cover to the pedestal.
15. Remove LOTO.
16. Restore system power.

4.2 Gear Box Assembly Replacement

Two people are required to replace the Gear Box Assembly due to its strong attraction to the magnet. The assembly must be removed from the Magnet Room prior to replacement. The Gear Box is also referred to as a Speed Reducer

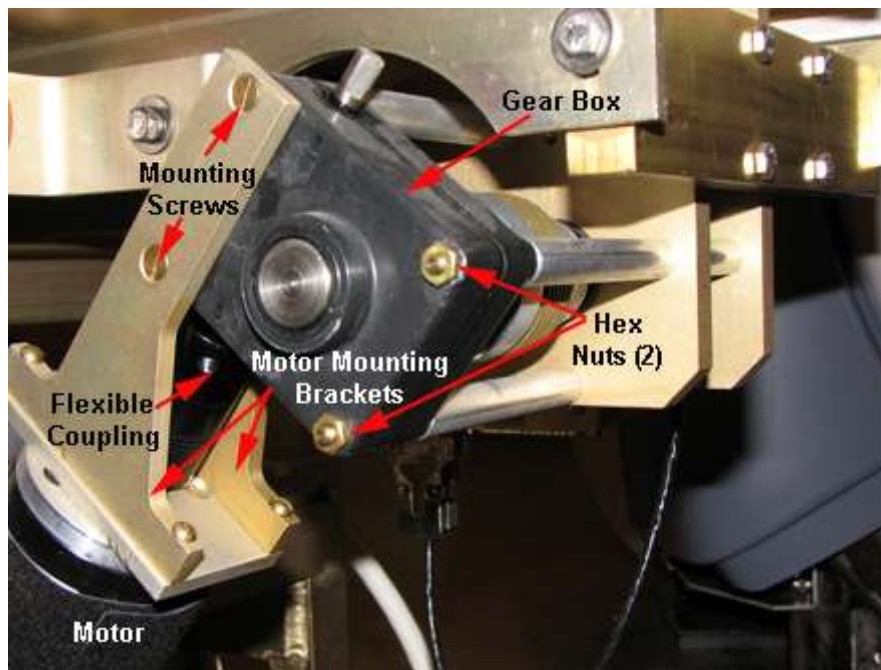
**CAUTION****Ferrous Material Hazard!**

Gear Box contains ferrous material.

Holding this assembly too close to the magnet bore will forcibly attract it to the magnet. To prevent possible bodily injury or material damage, remove the gear box from the Magnet Room.

1. Perform LOTO on the PDU. See the *MR Service Safety Manual*, PN 5452735.
2. Remove the rear pedestal left side cover (as viewed from the rear of the magnet).
3. Disconnect the cables from the motor.
4. Remove and retain the two hex nuts from the threaded rods that hold the motor/coupling/gear box assembly to the pedestal frame bracket.

Illustration 4: Mounted Drive Motor Assembly

**CAUTION**

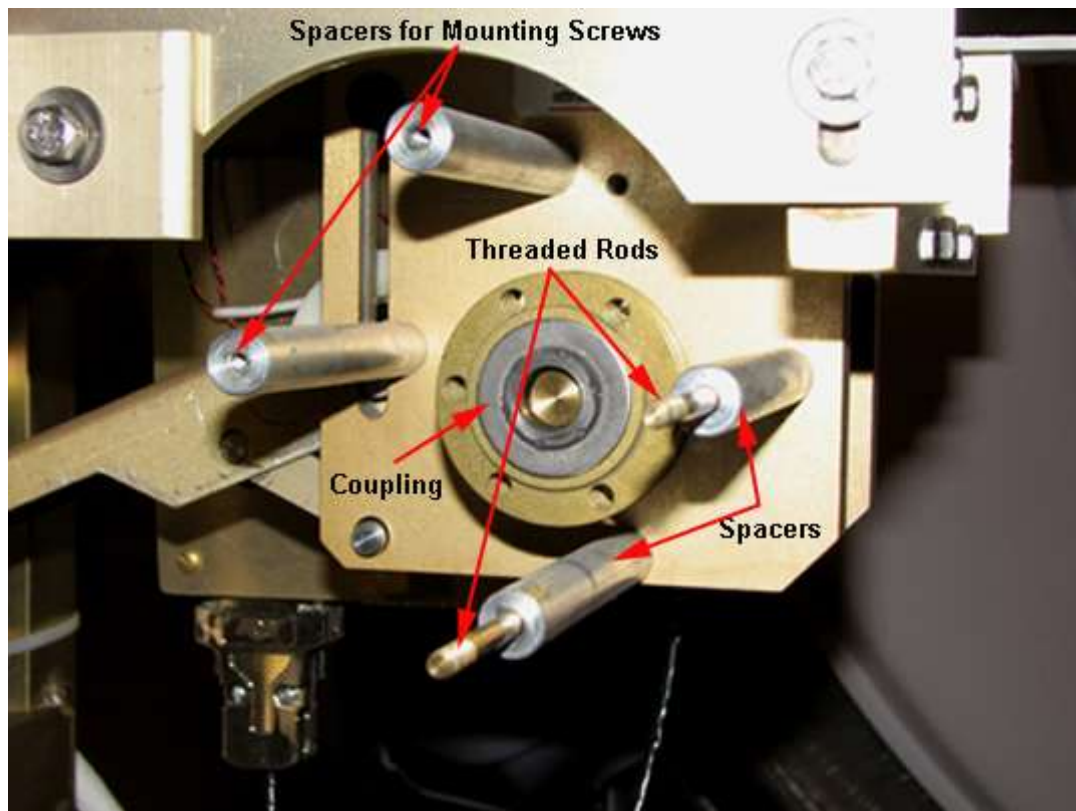
Ferrous Material Hazard!

Gear Box assembly contains ferrous material.

At no time should the gear box assembly be left within the magnetic field unattended. Once removal begins, the two people need to continue to carry the assembly until it is removed from the Magnet Room.

5. While one person holds the motor assembly firmly with both hands, the other person should loosen and remove the two long mounting screws that attach to the aluminum spacers. (Retain the screws since they will be used to reassemble the gear box to the motor bracket.)
6. Pull the assembly away from the mounts, spacers, and other coupling attached to the rear pedestal frame.

Illustration 5: Drive Motor Assembly Dismounted

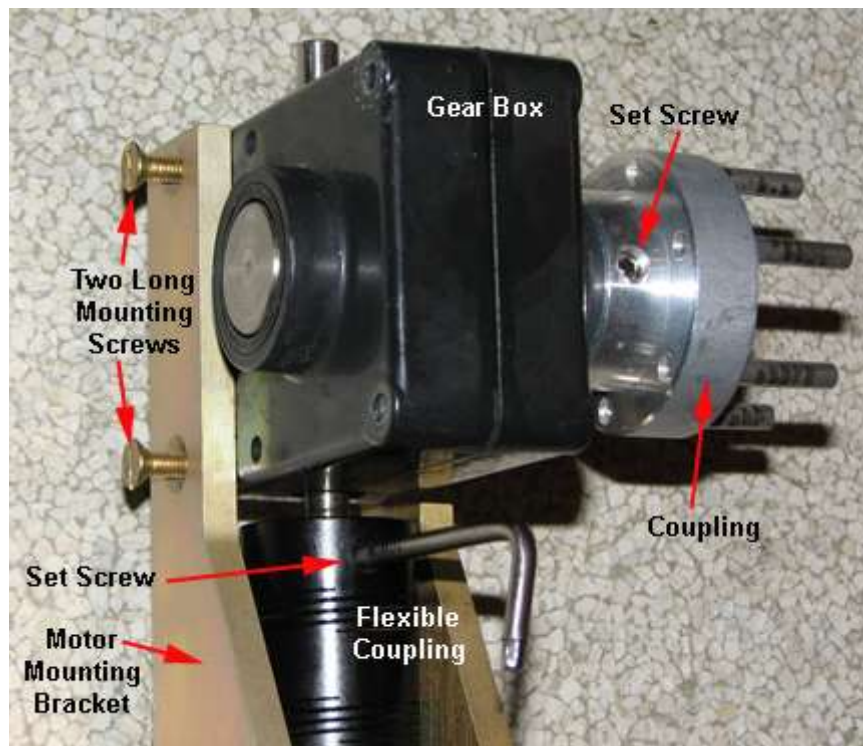
**NOTICE**

Use a towel or piece of cardboard to prevent damage to the floor and wall when removing the motor assembly.

7. Carefully remove the motor assembly from the Magnet Room by placing the assembly on the floor and against the wall farthest from the magnet, and slide the assembly to the front of the Magnet Room as shown in [Illustration 2](#).
8. With the motor/coupling/gear box assembly outside the Magnet Room, loosen the two set screws that are 90 degrees apart on the motor shaft flexible coupling (gear box side), and remove the Gear Box from the assembly. (Once removed, the gear box will no longer be held together.)

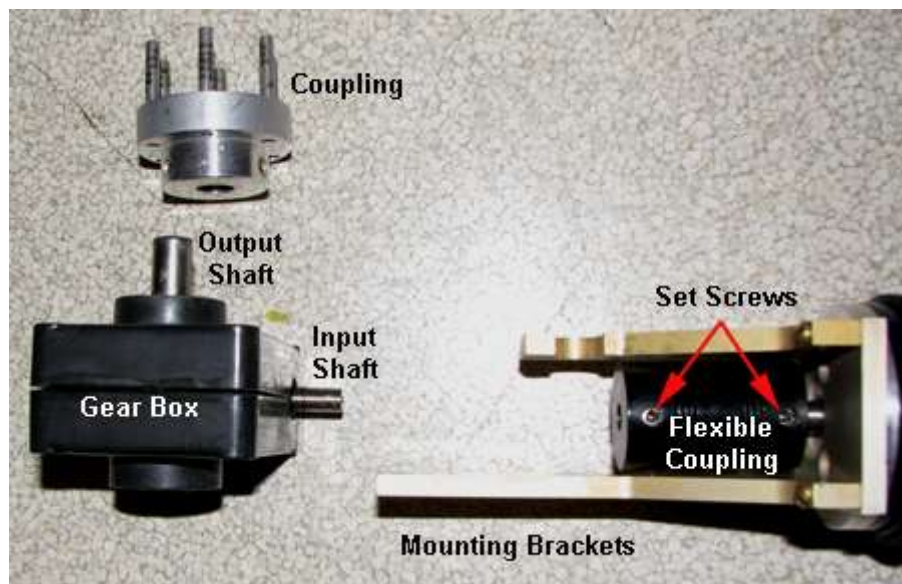
NOTE: There are four set screws on the flexible coupling at 90 degrees apart, two on each end.

Illustration 6: Set Screws on Drive Motor Assembly



9. Loosen the two set screws on the gear box output shaft coupling, and remove the coupling from the gear box.

Illustration 7: Disassembled Drive Motor Assembly



10. Prepare the new gear box for mounting by removing the four screws that hold the body together. (The gear box will utilize the mounting screws and threaded rod to hold it together in the drive motor assembly.)



NOTICE

Be sure to apply Loctite 242 to all screws used to reassemble the replacement assembly, including the set screws.

11. Attach the gear box output shaft coupling and tighten the two set screws.
12. Attach the gear box to the motor shaft flexible coupling by inserting the gear box in between the motor mounting brackets and sliding the gear box input shaft into the coupling.
- a. Use the two long mounting screws for positioning the gear box in the assembly.

- b. Ensure that one of the set screws is on the shaft flat and tighten the two set screws. (See [Illustration 6](#) for correct positioning of the components in the assembly.)

**CAUTION****Ferrous Material Hazard!**

Longitudinal Drive Motor contains ferrous material.

Holding this assembly too close to the magnet bore will forcibly attract it to the magnet. To prevent bodily injury or material damage, carefully bring the assembly into the Magnet Room along the same path used to remove it. (See [Illustration 2](#) for the proper process to remove the drive motor from the Magnet Room.) Two people are required at all times for this procedure.

13. While a second person is firmly holding the drive motor assembly, the other person should orient the assembly to the coupling, mounting screws and threaded rod, and reattach the assembly to the rear pedestal bracket as shown in [Illustration 5](#).
14. Tighten the two mounting screws and two hex nuts shown in [Illustration 4](#).
15. Reattach the cables to the motor and the cover to the rear pedestal.
16. Remove LOTO and restore system power.

4.3 Flexible Coupling Replacement

Follow either the Longitudinal Drive Motor or Gear Box replacement procedure to replace the flexible coupling.

- Be sure to loosen and tighten all four set screws (two are located at both ends) on the flexible coupling during replacement.
- Be sure to tighten one set screw each onto the motor shaft flat and one on the opposite end on the gear box input shaft flat. Apply Loctite 242 to all set screws.

5 Finalization

Ensure the cradle moves properly into the bore and out.

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